

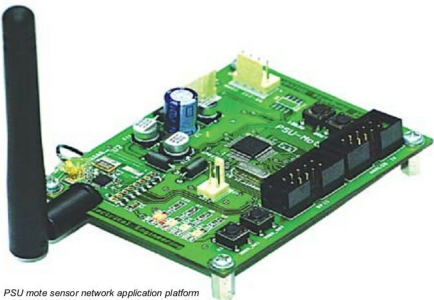
Part 1 of 2

Wireless Sensor Networks

Design Challenges and Embedded Platform Selection

While building a wireless sensor network, one faces several challenges that need to be addressed at the design stage itself. Moreover, as new wireless standards emerge, plenty of standards-based embedded products and solutions have become available to quickly build and deploy a wireless sensor network. Therefore selecting the right embedded product (sensor, processor, wireless transceiver) and its platform is a complex task.

This article explores ways to overcome the design challenges and gives tips on choosing the right embedded solution for an application



PSU mote sensor network: application platform

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A plethora of wireless sensor network (WSN) solutions are available today. These find applications ranging from toys and games, to consumer electronics and PC peripherals, home and building automation, healthcare, automotive, industrial and commercial. Here we look at the problems that the design-

ers face when implementing a wireless sensor network and the factors that are important in selecting a platform for deployment.

One major difficulty in platform selection is that different hardware/software solutions are difficult to compare, because there are no universally accepted metrics. It is hard to determine quantitatively which option is better than the other. The best ap-

proach is to rely on common sense and other guiding principles, such as open source versus proprietary code, or the open communication standard versus the one-vendor solution.

Primary design challenges

There are three key factors that impact the operation of a WSN: reliability, security and performance.

Reliability. For any WSN, the key goal is to have reliable data transmission, i.e., the highest probability of a packet being delivered in the presence of interference. Reliability refers to the packet error rate, error-correction coding, retransmission mechanisms and frequency planning. In the seven-layer OSI communication model, only the bottom four layers matter as far as reliability is concerned. Network topologies also affect the reliability of the communication system.

Security. Data protection in a WSN is a major challenge. Data security becomes even more significant when any security-oriented application is considered. Not only must the system ensure privacy, it must be able to authenticate data communication also. Security in-